
Pollity.in

Jan-Sunwai: Grievance Management Module

Product Requirements Document — MVP v0

Product Vision

Jan-Sunwai transforms how elected representatives manage constituent grievances — replacing scattered WhatsApp messages, paper registers, and missed follow-ups with a single, intelligent system that ensures every grievance is tracked, classified, assigned, and closed.

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Executive Summary

Jan-Sunwai is the first module of Pollity.in, an AI-empowered Digital Chief of Staff for India's 3.3 million elected representatives. This PRD defines the requirements for **MVP v0** — a working Phase 1 prototype to be demonstrated live at the SNVC Week 9 pitch in Spring 2026.

The prototype covers one use case end-to-end: a citizen sends a WhatsApp message, the system classifies it using AI, stores it, sends an acknowledgement, and displays it on a dashboard where office staff can track and update its status through closure.

This is a prototype, not a production system. MVP v0 is designed to validate the core value proposition with a single demo office. Production hardening, multi-tenancy, and full Auth happen in Phase 2 (post-SNVC).

Problem Statement

The Governance Gap

India's elected representatives — MPs, MLAs, municipal councillors, panchayat sarpanchs — receive hundreds of constituent grievances per week through a chaotic mix of channels: WhatsApp messages, in-person visits, phone calls, letters, and social media. There is no standard system for managing these.

The result is predictable:

- Grievances are forgotten or lost between intake and resolution
- No visibility into how many grievances exist, their status, or which departments are responsible
- Citizens receive no acknowledgement or follow-up, eroding trust
- Representatives cannot demonstrate accountability — they have no data on what they resolved
- Staff spend hours manually forwarding messages and chasing departments

The Specific Problem Jan-Sunwai Solves

Pain Point	What Jan-Sunwai Does
Grievances arrive on personal WhatsApp, get buried	Dedicated office number, every message captured
No categorisation or prioritisation	Claude AI classifies by category and urgency instantly
No acknowledgement to citizen	Automatic Ref ID sent back within seconds
No tracking after intake	Dashboard with full lifecycle from registered to closed
No accountability data	Every grievance timestamped, assigned, and searchable
Duplicate grievances from same citizen	AI duplicate detection flags repeat submissions

Why Now

WhatsApp Business API, capable AI classification at near-zero cost (Claude Haiku at \$0.00026 per grievance), and managed Postgres (Supabase free tier) make this buildable under \$500 total. This was not economically feasible two years ago.

Goals and Success Metrics

Phase 1 Goals (SNVC Prototype — 9 Weeks)

Goal	Description	Metric
G1	End-to-end flow works live	Demo completes without error at Week 9 pitch
G2	AI classification is accurate	$\geq 85\%$ correct category on test messages
G3	Acknowledgement is fast	Citizen receives Ref ID within 10 seconds
G4	Dashboard is usable	Non-technical staff can update a status in < 2 min
G5	System is stable	No crashes during the live demo

Phase 2 Goals (Post-SNVC — 20 Weeks)

Goal	Description	Metric
G6	Beta users onboarded	50 paying beta users
G7	Constituent satisfaction	Net Promoter Score > 40
G8	Grievance resolution rate	$\geq 70\%$ of registered grievances closed within 30 days
G9	Retention	$\geq 80\%$ of beta users active after 60 days
G10	Seed round unlock	Metrics sufficient for seed round conversation

What Does Not Count as Success

- Number of features built (irrelevant if G1–G5 are not met)
- Code quality metrics (prototype; refactor in Phase 2)
- Users outside the demo office (Phase 2 scope)

Users and Personas

Jan-Sunwai is **office-facing, not citizen-facing**. Citizens interact only through WhatsApp (intake and status updates). They never log in to the system.

Primary User: Office Staff

Who: Personal Assistant (PA) or Constituency Manager in the representative's office
Tech comfort: Uses WhatsApp daily; comfortable with Google Sheets; has never used a CRM or SaaS dashboard
Primary job: Receives grievances, forwards them to departments, follows up, reports back to the representative
Key frustration: Loses track of grievances after forwarding; citizens call back asking for status; has no way to show the representative what was done
What they need from Jan-Sunwai: See all open grievances in one place; update status in 3 clicks; know what needs action today

Secondary User: Elected Representative

Who: MLA, MP, municipal councillor, or panchayat sarpanch
Tech comfort: Uses WhatsApp; may not use a laptop regularly
Primary job: Constituent service, legislative work, political engagement
Key frustration: Cannot answer "how many grievances did we resolve this month?" Cannot hold staff accountable without data
What they need from Jan-Sunwai: A weekly summary of grievance resolution; a number they can quote publicly; confidence that nothing is falling through the cracks

Non-User: Citizen

The citizen is a **touchpoint**, not a user. They interact via WhatsApp only:

- **Input:** Sends a grievance message in any language
- **Output:** Receives an acknowledgement with a Reference ID
- They do not log in, do not see the dashboard, and do not have an account

User Stories

Priority notation: **MUST** = must have for v0 demo **SHOULD** = important but not blocking **COULD** = nice to have **WONT** = out of scope for v0

Citizen Stories

ID	Priority	Story
C1	MUST	As a citizen, I want to send a grievance via WhatsApp so that I can report a problem without visiting an office.
C2	MUST	As a citizen, I want to receive an acknowledgement with a Reference ID so that I know my grievance was received and can follow up.
C3	SHOULD	As a citizen, I want the acknowledgement to be in the same language I wrote in so that I can understand it.
C4	COULD	As a citizen, I want to receive a status update when my grievance is resolved so that I am informed of the outcome.
C5	WONT	As a citizen, I want to log in and track my grievance online. (Phase 2)

Office Staff Stories

ID	Priority	Story
S1	MUST	As office staff, I want to see all incoming grievances on a single dashboard so that I do not miss any.
S2	MUST	As office staff, I want each grievance to be automatically categorised and prioritised so that I know what to act on first.
S3	MUST	As office staff, I want to update the status of a grievance so that the system reflects the current state of resolution.
S4	MUST	As office staff, I want to assign a grievance to a department or person so that accountability is clear.
S5	MUST	As office staff, I want to see a summary of all open grievances so that I can report to the representative.
S6	SHOULD	As office staff, I want to filter grievances by category, urgency, and status so that I can focus on specific subsets.
S7	SHOULD	As office staff, I want to record a next action note on each grievance so that I remember what step is pending.
S8	COULD	As office staff, I want to see if a new grievance is a duplicate of an existing one so that I avoid processing the same issue twice.
S9	COULD	As office staff, I want to add walk-in grievances manually through the dashboard so that all channels are in one system.
S10	WONT	As office staff, I want to log in with a secure password. (Phase 2 — Auth)

Representative Stories

ID	Priority	Story
R1	SHOULD	As a representative, I want to see KPI metrics (total, open, critical, resolved today) so that I have an at-a-glance view of constituency health.
R2	COULD	As a representative, I want to see a breakdown of grievances by category so that I know which issues dominate my constituency.
R3	WONT	As a representative, I want a weekly email digest summarising grievance resolution. (Phase 2)

Functional Requirements

FR1 — WhatsApp Intake

ID	Pri	Requirement	Story
FR1.1	MUST	System must receive incoming WhatsApp messages via Meta Cloud API webhook	C1
FR1.2	MUST	System must validate every incoming webhook POST using HMAC-SHA256 signature	C1
FR1.3	MUST	System must respond to Meta's GET verification challenge	C1
FR1.4	MUST	System must process text messages; non-text messages (images, audio) are silently ignored in v0	C1
FR1.5	MUST	System must identify which office a message belongs to using the WhatsApp phone number ID	S1
FR1.6	MUST	Webhook must return HTTP 200 to Meta within 5 seconds regardless of processing time	C1

FR2 — AI Classification

ID	Pri	Requirement	Story
FR2.1	MUST	System must classify each grievance into one of 8 categories: infrastructure, welfare schemes, public safety, healthcare, education, land/revenue, corruption, others	S2
FR2.2	MUST	System must assign one of 4 urgency levels: critical, high, medium, low	S2
FR2.3	MUST	System must generate a 1–2 sentence English summary of the grievance	S5
FR2.4	MUST	System must detect the language of the incoming message (ISO 639-1 code)	C3
FR2.5	MUST	System must guard against prompt injection: citizen message text must be treated as data, never as instructions	S1
FR2.6	MUST	If the AI returns malformed output, the system must fall back to category=others, urgency=medium rather than crash	S1
FR2.7	SHOULD	System must check for duplicate grievances within the same office (same issue filed by the same citizen recently)	S8

FR3 — Grievance Storage

ID	Pri	Requirement	Story
FR3.1	MUST	Each grievance must be assigned a unique human-readable Reference ID in the format GR-{OFFICE}-{YEAR}-{SEQ}	C2
FR3.2	MUST	Reference ID sequence must be atomic — no two grievances for the same office can have the same ID	C2
FR3.3	MUST	System must store: raw text, category, urgency, summary, language, channel, citizen contact, office ID, timestamps	S1
FR3.4	MUST	System must store the full grievance lifecycle status	S3
FR3.5	SHOULD	System must store assigned_to and next_action fields	S4, S7
FR3.6	SHOULD	System must record closed_at timestamp when a grievance reaches closed status	S3

FR4 — Citizen Acknowledgement

ID	Pri	Requirement	Story
FR4.1	MUST	System must send an acknowledgement WhatsApp message to the citizen after registration	C2
FR4.2	MUST	Acknowledgement must include the Reference ID	C2
FR4.3	MUST	Acknowledgement for critical grievances must include an escalation notice	C2
FR4.4	SHOULD	Acknowledgement must include the grievance category so the citizen knows it was understood	C2

FR5 — Dashboard

ID	Pri	Requirement	Story
FR5.1	MUST	Dashboard must display a KPI row: total grievances, open, critical, resolved today	R1
FR5.2	MUST	Dashboard must display all grievances in a sortable, filterable table	S1
FR5.3	MUST	Table must show: Ref ID, date filed, urgency, category, status, summary, citizen contact, assigned to	S1
FR5.4	MUST	Dashboard must allow status updates: set new status, assigned_to, next_action	S3, S4
FR5.5	MUST	Dashboard must display a chart of open grievances by category	R2
FR5.6	MUST	Dashboard must display a chart of grievances by status	S5
FR5.7	SHOULD	Dashboard must support filtering by status, urgency, and category simultaneously	S6
FR5.8	SHOULD	Dashboard must auto-refresh data when a status update is submitted	S3
FR5.9	COULD	Dashboard must visually distinguish urgency levels by colour (critical=red, high=orange, medium=blue, low=green)	S2

FR6 — Grievance Lifecycle

The following status transitions are supported. Staff move grievances forward manually through the dashboard.

```
registered → acknowledged → assigned → in_progress → resolved → verified → closed
```

Status	Meaning
registered	Grievance received and classified. Ref ID issued to citizen.
acknowledged	Office has reviewed the grievance.
assigned	Forwarded to a department or staff member.
in_progress	Department has confirmed receipt and is working on it.
resolved	Department reports the issue is fixed.
verified	Office has independently confirmed resolution (not just paper).
closed	Case closed. Citizen may be notified (Phase 2).

Non-Functional Requirements

Performance

ID	Pri	Requirement
NF1.1	MUST	Webhook endpoint must return 200 to Meta within 5 seconds
NF1.2	MUST	End-to-end processing (webhook receipt to citizen acknowledgement) must complete within 15 seconds under normal load
NF1.3	SHOULD	Dashboard must load initial data within 3 seconds on a standard connection
NF1.4	COULD	System must handle up to 100 concurrent grievances per day without degradation (beta scale)

Reliability

ID	Pri	Requirement
NF2.1	MUST	Backend must not crash on malformed webhook payloads
NF2.2	MUST	Backend must not crash if Claude API is unavailable; must fall back gracefully
NF2.3	MUST	Railway deployment must stay alive 24/7 (no spin-down)
NF2.4	SHOULD	All grievance data must be backed up daily (Supabase handles this on free tier)

Security

ID	Pri	Requirement
NF3.1	MUST	All webhook POST requests must be validated with HMAC-SHA256 before processing
NF3.2	MUST	Service role key (Supabase) must never be exposed client-side
NF3.3	MUST	All secrets must be stored as environment variables, never hardcoded
NF3.4	MUST	Citizen message text must be treated as data (prompt injection guardrail)
NF3.5	SHOULD	Dashboard must use Supabase anon key with Row Level Security so staff only see their office's data
NF3.6	WONT	Full authentication with passwords and sessions (Phase 2)

Cost

ID	Pri	Requirement
NF4.1	MUST	Total Phase 1 infrastructure cost must not exceed \$85
NF4.2	MUST	Monthly recurring cost must not exceed \$7
NF4.3	MUST	AI cost (Claude Haiku) must not exceed \$1 for the Phase 1 demo period

Usability

ID	Pri	Requirement
NF5.1	MUST	A non-technical staff member must be able to update a grievance status in under 2 minutes with no training
NF5.2	SHOULD	Dashboard text must be in English for v0 (multilingual in Phase 2)
NF5.3	SHOULD	Dashboard must be accessible on a laptop browser without installation

Out of Scope for MVP v0

The following are explicitly **not** requirements for Phase 1. They are recorded here to prevent scope creep.

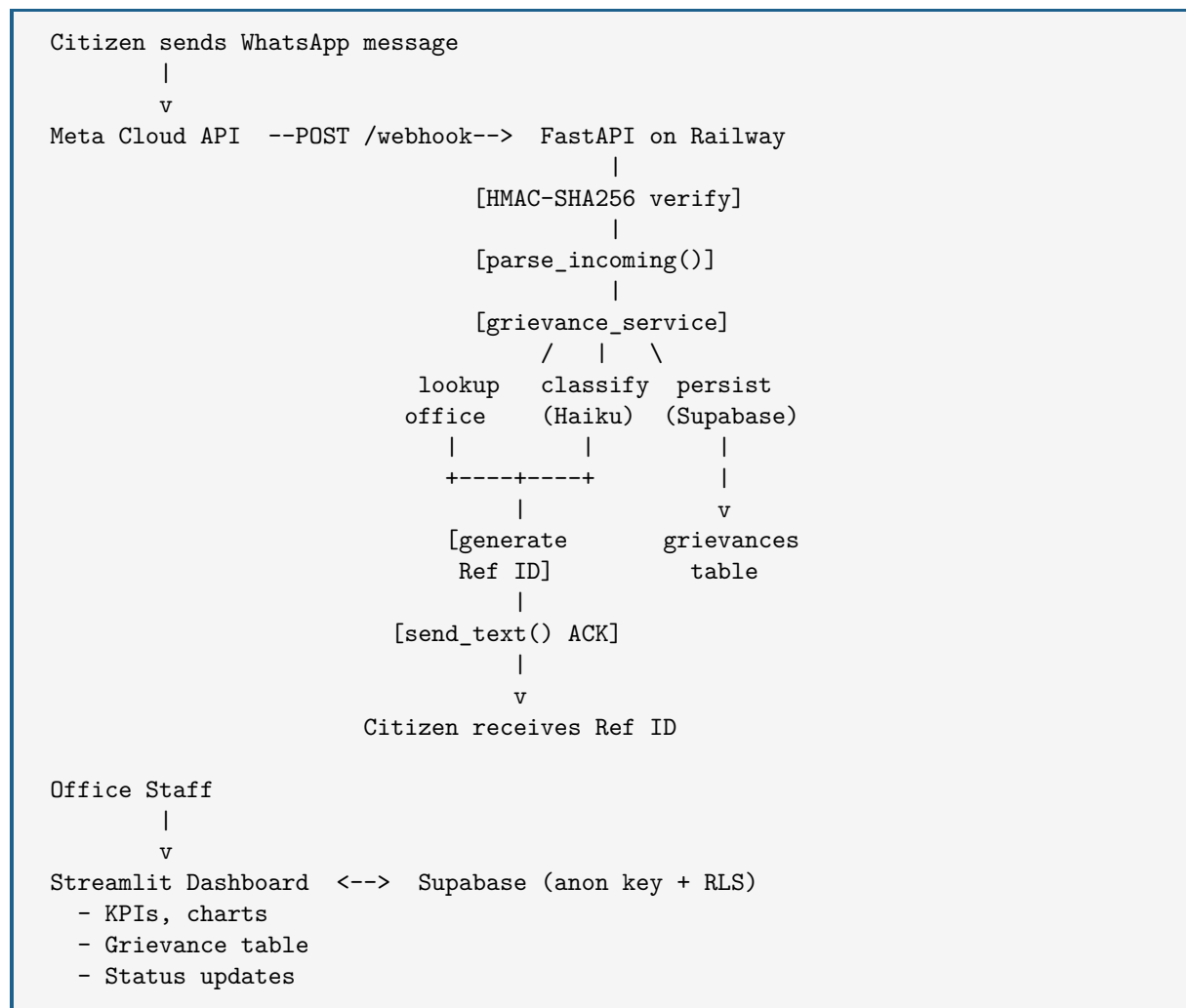
Feature	Reason deferred	Phase
Supabase Auth (login/password)	DEMO_OFFICE_ID env var used instead	Phase 2
Multi-tenancy (multiple offices)	Single demo office sufficient for SNVC pitch	Phase 2
Status updates sent to citizens	WhatsApp template approval takes time; not needed for demo	Phase 2
Walk-in / phone intake form	Dashboard form; not needed for WhatsApp demo	Phase 2
Weekly digest email	Resend integration; out of demo scope	Phase 2
CPGRAMS integration	Complex API; not needed for prototype	Phase 2
Pattern recognition reports	Requires sufficient data volume	Phase 2
Referral letter drafting	Valuable but not core to demo flow	Phase 2
Media messages (images, audio)	Text-only classification sufficient for demo	Phase 2
Mobile-optimised dashboard	Laptop browser sufficient for demo	Phase 2
DPDP Act compliance	Railway on US servers; formal compliance post-seed	Post-seed
Vidhayak-Mitra module	Legislative research module; separate build	Post-seed
Vikas-Tracker module	Fund management module; separate build	Post-seed
Nagrik-Setu module	Social media / engagement module; separate build	Post-seed

System Architecture Summary

Components

Component	Technology	Role	Cost
Backend API	FastAPI (Python 3.12)	Webhook handler, classifier orchestration, CRUD API	—
Hosting	Railway	Keeps backend alive 24/7, public HTTPS URL	\$5/mo
Database	Supabase (Postgres)	Grievance storage, RLS, atomic counter RPC	Free
AI Classifier	Claude Haiku	Category, urgency, summary, language, duplicate	~\$0.13
Messaging	Meta WhatsApp Cloud	Intake webhook, outbound ACK	Free
Dashboard	Streamlit	Staff-facing UI, charts, status updates	Free
CI/CD	GitHub Actions	Import checks, unit tests on push	Free

Data Flow



Acceptance Criteria for Demo (SNVC Week 9)

The following scenarios must work live at the pitch without failure.

ID	Status	Scenario
AC1	☐	Send a Hindi infrastructure grievance via WhatsApp → citizen receives Ref ID within 15 seconds
AC2	☐	Dashboard shows the new grievance with correct category (infrastructure) and urgency
AC3	☐	Send a critical safety grievance → urgency = critical, ACK contains escalation notice
AC4	☐	Update grievance status from registered to assigned, add assigned_to → dashboard reflects change
AC5	☐	Dashboard KPI row shows correct counts (total, open, critical)
AC6	☐	Filter grievances by category → only matching rows displayed
AC7	☐	Close a grievance → removed from open count, closed_at timestamp recorded
AC8	☐	Send a tampered webhook (bad HMAC) → system returns 401, no grievance created

AC8 is for internal testing only — do not demo the security rejection to judges unless asked.
Demo AC1–AC7 in sequence for maximum narrative impact.

Risks and Mitigations

Description WhatsApp Business API approval takes longer than expected; production number not ready before demo.

Mitigation Apply on Day 1. Use Meta's built-in test number for the SNVC demo — it is fully functional and requires no approval.

Description Backend is unreachable at the moment of the live pitch due to a Railway incident or deployment error.

Mitigation Run a full end-to-end test 24 hours before the pitch. Record a screen capture of the working demo as a backup.

Description Anthropic's Claude Haiku API is unavailable during the demo, blocking classification and blocking the ACK.

Mitigation Add a hardcoded fallback: if Haiku times out, classify as **others / medium** and still issue the Ref ID. Demo is not blocked.

Description Supabase free tier caps at 500 MB storage and 50,000 rows. Could be exceeded with heavy use.

Mitigation Not reachable at demo scale (tens of grievances). No action needed for Phase 1. Monitor in Phase 2.

Description Live demo fails mid-pitch due to slow or unreliable Wi-Fi at the SNVC venue.

Mitigation Pre-load the dashboard before going on stage. Use phone hotspot as backup. Have a screen recording ready as last resort.

Description	Team is tempted to add Phase 2 features (Auth, multi-tenancy, email digests) to Phase 1, delaying the demo-ready build.
Mitigation	This PRD is the contract. Section 8 (Out of Scope) is the definitive list. Any addition requires explicit agreement from both founders.

Phase 1 Build Milestones

Week	SNVC Deliverable	Product Build Target
1	Problem framing	Repo scaffold, Supabase schema live
2	Customer discovery	WhatsApp webhook receiving messages
3	Customer interviews	Claude classifier integrated, ACK sending
4	Value proposition	Dashboard v0 deployed on Streamlit Cloud
5	Business model	End-to-end flow working (AC1–AC3 passing)
6	Competitive analysis	Dashboard filters, status updates (AC4–AC7)
7	Financial model	All acceptance criteria passing
8	Pitch rehearsal	Bug fixes, demo polish, backup recording
9	Final Pitch	Live demo at SNVC

Open Questions

The following questions are unresolved as of this version of the PRD. They must be answered before Phase 2 planning.

1. **Pricing:** What is the right monthly price for a beta user? Options under consideration: \$0 (free beta) / \$10 / \$25 per office per month.
2. **Language:** Should the acknowledgement message to citizens be sent in the same language as their message? Requires Claude to translate the ACK — adds latency and cost.
3. **Phone number:** Will each office get their own WhatsApp number, or share one? Separate numbers are cleaner for citizens but require more Meta WABA management.
4. **Onboarding:** How do we onboard 50 beta offices post-SNVC? Manual white-glove, or self-serve?
5. **Indian data residency:** DPDP Act requires data localisation. Railway currently runs on US/EU servers. Do we need to host on AWS Mumbai before Phase 2 beta, or after seed?
6. **WhatsApp template messages:** For status updates to citizens (Phase 2), Meta requires pre-approved message templates. When do we apply for approval — during Phase 1 or at start of Phase 2?